LITERATURE REVIEW

Double/Debiased Machine Learning for Treatment and Structural Parameters

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ANALYTICAL ABSTRACT

The paper asks how to draw valid root-N inference on a low-dimensional parameter, typically a causal effect, when the nuisance functions needed to identify it are so complex that they must be estimated by machine learning. Plugging a regularised learner into the natural estimating equation transmits first-order regularisation bias into the estimate, and using the same observations to fit and to evaluate the learner adds an overfitting bias. The authors remove the first by moving to a Neyman-orthogonal score, whose pathwise derivative with respect to the nuisance parameters vanishes at the truth, and the second by cross-fitting. Under a rate condition on the product of the nuisance estimation errors, the resulting double or debiased machine learning estimator is asymptotically normal and uniformly valid over a class of distributions. The theory is applied to partially linear regression and instrumental variable models, to average and local average treatment effects, and illustrated on a reemployment bonus experiment, on 401(k) eligibility and on settler mortality and institutions. The review reconstructs the bias decomposition and the main proofs, evaluates the product-rate condition, and argues that every conclusion is conditional on unconfoundedness, overlap, independent sampling and sufficiently accurate learners, none of which the method can itself test.

1 Introduction

1.1 Paper identification and classification

The paper is Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey and Robins (2018), published in *The Econometrics Journal* 21(1), pages C1 to C68, DOI 10.1111/ectj.12097, and circulated earlier as arXiv 1608.00060 and as NBER Working Paper 23564. It is an econometric and methodological paper with a general asymptotic theory, three specialisations to named models and three empirical illustrations. The field is econometrics and the subfield is semiparametric estimation and causal inference with high-dimensional nuisance parameters. The version read in full for this review is the complete text of arXiv 1608.00060 as rendered in HTML, including the main text, the empirical tables, the footnotes and the appendix with the proofs of the lemmas and of the first theorem. The journal PDF could not be downloaded in the environment used for the review, so no copy of the paper is stored beside this report.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

The paper looks for a general recipe that delivers a point estimate of a finite-dimensional parameter θ_0 that is consistent at rate $N^{-1/2}$, asymptotically normal, and accompanied by honest confidence intervals, when the nuisance functions η_0 entering the identification of θ_0 are estimated by lasso, random forests, boosted trees, neural networks or ensembles of these. The target is not a better prediction of the outcome but a valid statement about one coefficient or one treatment effect.

WHY DOES THE QUESTION MATTER?

Machine learning methods predict well because they regularise, and regularisation induces bias. If the quantity of interest is a causal coefficient, that bias does not wash out: it enters the coefficient of interest at order larger than $N^{-1/2}$. The practical consequence is that the standard applied workflow, which is to control for many covariates flexibly and read off a coefficient, has no inferential guarantee when the controls are selected or fitted adaptively. The question matters for any field that relies on selection on observables, which includes household finance, corporate finance and empirical asset pricing.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Classical semiparametric theory, from Neyman (1959) through Robinson (1988), Andrews (1994), Newey (1994) and van der Vaart (1998), shows that root-N inference is possible when the nuisance estimator lies in a Donsker class. That condition fails once the dimension of the covariates grows with the sample size. Belloni, Chernozhukov and Hansen and several co-authors showed in a series of papers that the same orthogonality idea combined with lasso gives valid inference under sparsity, but the proofs rely on restrictive entropy growth conditions. What was missing was a

general statement that accommodates non-sparse nuisance structure and arbitrary learners, with the learner's quality summarised by a single convergence rate.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

The paper proposes the double or debiased machine learning estimator, which has two ingredients: a Neyman-orthogonal estimating equation and cross-fitting. It investigates the estimator by deriving a general theorem for linear and nonlinear orthogonal scores, specialising it to four models, and checking finite-sample behaviour in two simulations and three applications. It finds that the estimator is asymptotically normal and uniformly valid when the product of nuisance estimation errors is $o(N^{-1/2})$, and that in the three applications the results are stable across five learners and two hybrids.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The question is whether, for a parameter θ_0 defined by a moment condition $\mathrm{E}[\psi(W;\theta_0,\eta_0)] = 0$ with an infinite-dimensional or high-dimensional nuisance η_0 that is estimated by a machine learner at a rate slower than $N^{-1/2}$, there is a construction of the score ψ and of the estimation sample such that $\sqrt{N}(\tilde{\theta}_0 - \theta_0)$ is asymptotically normal, uniformly over a large set of data generating processes, without Donsker conditions on the learner.

2.2 Hypotheses

- The naive plug-in estimator built from a non-orthogonal score is not $\sqrt{N}$-consistent, because regularisation bias in $\hat{g}_0$ contaminates it at order $n^{-\varphi_g}$ with $\varphi_g < 1/2$.
- Replacing the score by a Neyman-orthogonal one makes the bias depend on a product of two nuisance errors, which can be $o(N^{-1/2})$ even when each error is only $o(N^{-1/4})$.
- Estimating the nuisance on one part of the sample and evaluating the score on another part removes the overfitting bias for any learner, with no complexity restriction, and cross-fitting recovers full efficiency.
- Consequently the estimate should be insensitive to which sensible learner is used, which the applications are designed to check.

2.3 Existing literature and research gap

PAPER The authors place their work between two literatures: root-N estimation of low-dimensional objects with nonparametric nuisances (Levit, Ibragimov and Hasminskii, Bickel, Robinson, Newey, Andrews, Robins and Rotnitzky, van der Laan and Rose), and the use of sample splitting to relax entropy conditions (Bickel 1982, Schick 1986, Angrist and Krueger 1995, Belloni et al. 2010 and 2012, Robins et al. 2013). They state that existing high-dimensional results require, for example, sparsity of order $s \ll \sqrt{n}$ for the optimal instrument without splitting.

CRITIQUE The gap is real but narrower than the introduction suggests. Orthogonal scores for treatment effects (the augmented inverse propensity weighted score) already appear in Robins and Rotnitzky (1995) and Hahn (1998), and sample splitting for nuisance estimation was already used in the targeted learning literature. The new element is the packaging: a single set of high-level conditions under which any learner meeting a rate bound yields uniform inference, and the demonstration that cross-fitting rather than a one-off split is what preserves efficiency.

2.4 Claimed contribution

CRITIQUE The claimed contribution is a general, simple theory for inference in the presence of machine-learned nuisances. It is demonstrated formally, since Theorems 3.1 and 3.3 are proved from stated assumptions and the proof structure is transparent. It is demonstrated less well as a practical claim that the method is robust to the choice of learner, because the three applications are not accompanied by any validation of the nuisance rate condition, and the simulations cover one data generating process each.

3 Theoretical Framework and Mechanism

3.1 Why the naive estimator fails: regularisation bias

PAPER The lead example is the partially linear regression of Robinson (1988):

$$Y = D\theta_0 + g_0(X) + U, \quad \mathbf{E}[U \mid X, D] = 0, \qquad D = m_0(X) + V, \quad \mathbf{E}[V \mid X] = 0. \tag{1}$$

Y	Outcome variable.
D	Scalar treatment or policy variable whose coefficient is the target.
X	Vector of p controls, with p allowed to grow with the sample size.
$g_0(X)$	Effect of the controls on the outcome, a nonparametric function.
$m_0(X)$	Conditional mean of the treatment given the controls, which records confounding.
U, V	Disturbances; V is the part of treatment that is not predictable from X.

The first equation defines θ_0 and the second keeps track of how X pushes on D. If D is as good as randomly assigned given X, then θ_0 is the average treatment effect. The vector $\eta_0 = (m_0, g_0)$ is the nuisance.

Suppose the sample is split into a main part I of size n and an auxiliary part, $\hat{g}_0$ is learned on the auxiliary part, and the natural estimator is formed on the main part:

$$\hat{\theta}_0 = \Big(\frac{1}{n}\sum_{i\in I} D_i^2\Big)^{-1} \frac{1}{n}\sum_{i\in I} D_i\big(Y_i - \hat{g}_0(X_i)\big). \tag{2}$$

Substituting $Y_i = D_i\theta_0 + g_0(X_i) + U_i$ gives the exact decomposition

$$\begin{aligned} \sqrt{n}(\hat{\theta}_0 - \theta_0) = a + b, \qquad a = \Big(\tfrac{1}{n}\sum_{i\in I} D_i^2\Big)^{-1} \tfrac{1}{\sqrt{n}}\sum_{i\in I} D_i U_i, \\ b = \Big(\tfrac{1}{n}\sum_{i\in I} D_i^2\Big)^{-1} \tfrac{1}{\sqrt{n}}\sum_{i\in I} D_i\big(g_0(X_i) - \hat{g}_0(X_i)\big). \end{aligned} \tag{3}$$

Term a is a normalised sum of mean-zero terms and converges to a normal law. Term b is the problem. Because $D_i = m_0(X_i) + V_i$ and V_i is mean zero given X_i, term b equals, to first order, $(\mathrm{E}D^2)^{-1} n^{-1/2} \sum_{i\in I} m_0(X_i)\,(g_0(X_i) - \hat{g}_0(X_i))$. These summands do not have mean zero, since a regularised learner has a systematic error. If $\hat{g}_0$ converges at rate $n^{-\varphi_g}$ with $\varphi_g < 1/2$, which is what regularisation buys for rich nuisance classes, then b is of order $\sqrt{n}\, n^{-\varphi_g} \to \infty$. So $|\sqrt{n}(\hat{\theta}_0 - \theta_0)| \to_p \infty$: the estimator is not root-n consistent and the normal approximation is wrong. The intuition is that the score $D(Y - D\theta - g(X))$ treats a mistake in g as a first-order mistake in θ, because D is correlated with g through m_0.

3.2 Orthogonalisation: the product structure

PAPER The remedy is to partial out X from D as well. Let $\hat{V} = D - \hat{m}_0(X)$ with $\hat{m}_0$ learned on the auxiliary sample. The debiased estimator is

$$\check{\theta}_0 = \Big(\frac{1}{n}\sum_{i\in I} \hat{V}_i D_i\Big)^{-1} \frac{1}{n}\sum_{i\in I} \hat{V}_i\big(Y_i - \hat{g}_0(X_i)\big), \tag{4}$$

which is a linear instrumental variable estimator with the residual $\hat{V}$ as instrument. Its scaled error splits into three pieces,

$$\begin{aligned} \sqrt{n}(\check{\theta}_0 - \theta_0) = A_n + B_n + C_n, \\ B_n = (\mathrm{E}V^2)^{-1} \frac{1}{\sqrt{n}}\sum_{i\in I} \big(\hat{m}_0(X_i) - m_0(X_i)\big)\big(\hat{g}_0(X_i) - g_0(X_i)\big), \end{aligned} \tag{5}$$

where $A_n = (\mathrm{E}V^2)^{-1} n^{-1/2} \sum_{i\in I} V_i U_i$ is the leading normal term and C_n is a remainder containing terms such as $n^{-1/2} \sum_{i\in I} V_i(\hat{g}_0(X_i) - g_0(X_i))$. The bias term B_n now depends on the product of the two estimation errors. If $\hat{m}_0$ and $\hat{g}_0$ converge at rates $n^{-\varphi_m}$ and $n^{-\varphi_g}$, then $|B_n| \lesssim \sqrt{n}\, n^{-(\varphi_m + \varphi_g)}$, which vanishes whenever

$\varphi_m + \varphi_g > 1/2$. Each learner may therefore be slower than $n^{-1/4}$ in one direction as long as the other compensates.

INFERENCE The same property can be verified exactly, without asymptotics, which the paper only asserts. Write the orthogonal score as $\psi(W;\theta,\eta) = (Y - D\theta - g(X))(D - m(X))$. At θ_0, $Y - D\theta_0 - g(X) = U + (g_0 - g)(X)$ and $D - m(X) = V + (m_0 - m)(X)$. Expanding the product and using $\mathrm{E}[UV] = 0$, $\mathrm{E}[U(m_0 - m)] = 0$ (because $\mathrm{E}[U \mid X, D] = 0$) and $\mathrm{E}[V(g_0 - g)] = 0$ (because $\mathrm{E}[V \mid X] = 0$) leaves

$$\mathrm{E}\,\psi(W;\theta_0,g,m) = \mathrm{E}\big[(g_0(X) - g(X))(m_0(X) - m(X))\big]. \tag{6}$$

The expected score at the true θ_0 is exactly the covariance of the two nuisance errors. Setting $g = g_0 + r(\bar{g} - g_0)$ and $m = m_0 + r(\bar{m} - m_0)$ gives $r^2\mathrm{E}[(\bar{g} - g_0)(\bar{m} - m_0)]$, whose first derivative at $r = 0$ is zero (Neyman orthogonality) and whose second derivative is $2\mathrm{E}[(\bar{g} - g_0)(\bar{m} - m_0)]$, the quantity the paper calls λ'_N. By contrast, the non-orthogonal score $\varphi = (Y - D\theta - g(X))D$ gives $\mathrm{E}\varphi(W;\theta_0,g) = \mathrm{E}[(g_0 - g)(X)\, m_0(X)]$, which is linear in the error and does not vanish unless $m_0 \equiv 0$. This is the whole mechanism in two lines.

3.3 Neyman orthogonality and its constructions

PAPER The general moment condition is $\mathrm{E}_P[\psi(W;\theta_0,\eta_0)] = 0$. Define the Gateaux derivative map $\mathrm{D}_r[\eta - \eta_0] = \partial_r \mathrm{E}_P[\psi(W;\theta_0,\eta_0 + r(\eta - \eta_0))]$ for $r \in [0,1)$. Definition 2.1 requires

$$\partial_\eta \mathrm{E}_P\,\psi(W;\theta_0,\eta_0)[\eta - \eta_0] = 0 \quad \text{for all } \eta \in \mathcal{T}_N, \tag{7}$$

where $\mathcal{T}_N$ is a shrinking neighbourhood of η_0 that contains $\hat{\eta}_0$ with high probability. Definition 2.2 relaxes this to near-orthogonality, $\|\partial_\eta \mathrm{E}_P\psi[\eta - \eta_0]\| \le \lambda_N = o(N^{-1/2})$. The condition says that the moment equation identifying θ_0 is locally insensitive to the nuisance value, so noisy nuisance estimates barely move the solution.

The paper offers several recipes for producing such scores from a non-orthogonal one.

PAPER The first recipe, Neyman's, applies when (θ_0,β_0) maximise $\mathrm{E}_P\ell(W;\theta,\beta)$ for a criterion ℓ. Set

$$\psi(W;\theta,\eta) = \partial_\theta \ell(W;\theta,\beta) - \mu\,\partial_\beta \ell(W;\theta,\beta), \qquad \mu_0 = J_{\theta\beta}J_{\beta\beta}^{-1}, \tag{8}$$

ℓ	Criterion or quasi log-likelihood.
β	Finite-dimensional nuisance parameter.
μ	Orthogonalisation matrix of size $d_\theta \times d_\beta$, itself a nuisance.
J_{ab}	Blocks of the expected Hessian $\partial_{(\theta,\beta)}\mathrm{E}[\partial_{(\theta,\beta)}\ell]$ at the truth.

The score subtracts from the θ-score its projection on the β-score. Orthogonality follows because $\partial_\eta \mathrm{E}\psi = ([J_{\theta\beta} - \mu_0 J_{\beta\beta}],\ \mathrm{E}[\partial_\beta \ell] \otimes I)$, and the first block vanishes by the choice of μ_0 while the second vanishes by the first-order condition $\mathrm{E}[\partial_\beta \ell] = 0$ (Lemma 2.1). In the linear regression case $\ell = -(Y - D\theta - X'\beta)^2/2$, this returns $\psi = (Y - D\theta - X'\beta)(D - \gamma_0' X)$, the partialling-out score; if $\mathrm{E}[XX']$ is ill-conditioned, μ_0 is replaced by the regularised solution of a relaxed constraint $\|J_{\theta\beta} - \mu J_{\beta\beta}\|_q \le r_N$, which delivers near-orthogonality at the cost of some efficiency (Lemma 2.2).

PAPER The second recipe, concentrating out, extends this to infinite-dimensional β. With $\beta_\theta = \arg\max_\beta \mathrm{E}_P \ell(W;\theta,\beta)$, the score is the total derivative

$$\psi(W;\theta,\eta) = \frac{d\,\ell(W;\theta,\eta(\theta))}{d\theta}, \qquad \eta_0(\theta) = \beta_\theta. \tag{9}$$

Orthogonality holds because the derivative of $\mathrm{E}\ell$ in any direction of β is zero at the maximiser β_θ for every θ, and mixed partial derivatives commute (Lemma 2.5, whose proof is in the appendix). For the partially linear model, $\ell = -(Y - D\theta - \beta(X))^2/2$ gives $\beta_\theta(X) = \mathrm{E}[Y - D\theta \mid X]$ and

$$\psi = (D - \mathrm{E}[D \mid X])\big(Y - \mathrm{E}[Y \mid X] - (D - \mathrm{E}[D \mid X])\theta\big) = (D - m_0(X))\big(Y - D\theta - g_0(X)\big), \tag{10}$$

the score of Equation (4). A mean-squared-error criterion is enough; no likelihood is needed.

PAPER The remaining recipes cover GMM, $\psi = \mu\, m(W;\theta,\beta)$ with $\mu_0 = A'\Omega^{-1} - A'\Omega^{-1}G_\beta(G_\beta'\Omega^{-1}G_\beta)^{-1}G_\beta'\Omega^{-1}$ (Lemma 2.3), conditional moment restrictions with functional nuisances (Lemma 2.6), and the influence function adjustment of Newey (1994), $\psi = \varphi + \phi$, where ϕ corrects the original score for the first-step nonparametric estimation (Equation 2.30). In the partially linear model the adjustment is $\phi = -m_0(X)\{Y - D\theta - \beta(X,\theta)\}$, which converts $D(Y - D\theta - g)$ into the orthogonal score. The recipes are equivalent routes to the same object, and the choice of the weighting matrix Ω and selection matrix A only matters for efficiency.

3.4 Cross-fitting and the DML estimators

PAPER Orthogonality controls the bias term B_n but not the remainder C_n, which contains products of structural noise and the learner's error evaluated on the same observations the learner saw. If $\hat{g}_0$ is estimated on observations outside I, then conditional on the auxiliary sample $\hat{g}_0$ is fixed, and $n^{-1/2}\sum_{i\in I} V_i(\hat{g}_0 - g_0)(X_i)$ has mean zero and variance $n^{-1}\sum_{i\in I}(\hat{g}_0 - g_0)^2(X_i) \to_P 0$, so it vanishes by Chebyshev's inequality. No entropy restriction on the learner is needed. The paper's own cautionary example is $\hat{g}_0(X_i) = g_0(X_i) + (Y_i - g_0(X_i))/N^{1/2-\kappa}$ for observations inside the estimation sample. This estimator converges uniformly at nearly the parametric rate $N^{-1/2+\kappa}$, yet the remainder equals

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N} V_i\big(\hat{g}_0(X_i) - g_0(X_i)\big) \;\propto\; N^{\kappa} \to \infty \tag{11}$$

when the full sample is used for both fitting and evaluation.

PAPER Definitions 3.3 and 3.4 formalise the cross-fitted estimators. Take a random partition $(I_k)_{k=1}^K$ of $\{1,\dots,N\}$ into folds of size $n = N/K$, let $\hat{\eta}_{0,k}$ be estimated on I_k^c, and let $\mathbb{E}_{n,k}$ be the average over fold k. DML1 solves the score equation fold by fold and averages, and DML2 pools the folds,

$$\text{DML1: } \mathbb{E}_{n,k}\big[\psi(W;\check{\theta}_{0,k},\hat{\eta}_{0,k})\big] = 0,\;\; \tilde{\theta}_0 = \tfrac{1}{K}\sum_{k=1}^{K}\check{\theta}_{0,k}; \qquad \text{DML2: } \tfrac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{n,k}\big[\psi(W;\tilde{\theta}_0,\hat{\eta}_{0,k})\big] = 0. \tag{12}$$

Both are first-order equivalent. The authors recommend DML2, because the pooled empirical Jacobian is more stable than the fold-specific ones, and moderate K such as 4 or 5 rather than 2,

because the learner then sees $(K-1)/K$ of the data. To reduce dependence on the random partition they repeat the procedure S times and report the median estimate $\tilde{\theta}_0^{\text{median}}$ together with a variance that adds the dispersion across splits,

$$\hat{\sigma}^{2,\text{median}} = \text{median}_s \left\{ \hat{\sigma}_s^2 + (\hat{\theta}_s - \tilde{\theta}^{\text{median}})(\hat{\theta}_s - \tilde{\theta}^{\text{median}})' \right\}. \tag{13}$$

3.5 The main theorem for linear scores

PAPER For scores that are linear in θ, $\psi(w;\theta,\eta) = \psi^a(w;\eta)\theta + \psi^b(w;\eta)$, Assumption 3.1 requires the moment condition, twice continuous Gateaux differentiability, orthogonality with $\lambda_N \le \delta_N N^{-1/2}$, and an identification condition that the singular values of $J_0 = \mathrm{E}\psi^a(W;\eta_0)$ lie in $[c_0, c_1]$. Assumption 3.2 requires that $\hat{\eta}_0$ lies in the realisation set $\mathcal{T}_N$ with probability at least $1-\Delta_N$, with moment bounds and three statistical rates,

$$\begin{aligned} r_N &= \sup_{\eta \in \mathcal{T}_N} \left\| \mathrm{E}\psi^a(W;\eta) - \mathrm{E}\psi^a(W;\eta_0) \right\|, \\ r'_N &= \sup_{\eta \in \mathcal{T}_N} \left(\mathrm{E}\|\psi(W;\theta_0,\eta) - \psi(W;\theta_0,\eta_0)\|^2 \right)^{1/2}, \\ \lambda'_N &= \sup_{r,\eta} \left\| \partial_r^2 \mathrm{E}\psi(W;\theta_0,\eta_0 + r(\eta-\eta_0)) \right\|, \end{aligned} \tag{14}$$

each bounded by δ_N, with $\lambda'_N \le \delta_N/\sqrt{N}$. Theorem 3.1 then states that, uniformly over $P \in \mathcal{P}_N$,

$$\begin{aligned} \sqrt{N}\sigma^{-1}(\tilde{\theta}_0 - \theta_0) &= \frac{1}{\sqrt{N}} \sum_{i=1}^N \bar{\psi}(W_i) + O_P(\rho_N) \xrightarrow{d} N(0, \mathrm{I}_d), \\ \rho_N &= N^{-1/2} + r_N + r'_N + N^{1/2}\lambda_N + N^{1/2}\lambda'_N, \end{aligned} \tag{15}$$

$\bar{\psi}$	The influence function $-\sigma^{-1}J_0^{-1}\psi(\cdot;\theta_0,\eta_0)$.
σ^2	Asymptotic variance $J_0^{-1}\mathrm{E}[\psi\psi'](J_0^{-1})'$ evaluated at the truth.
ρ_N	Size of the remainder; the theorem is meaningful when $\rho_N \to 0$.

and Theorem 3.2 shows that the plug-in variance $\hat{\sigma}^2 = \hat{J}_0^{-1}\frac{1}{K}\sum_k \mathbb{E}_{n,k}[\psi\psi']\hat{J}_0^{-1\prime}$ is consistent, so that the interval $\ell'\tilde{\theta}_0 \pm \Phi^{-1}(1-\alpha/2)\sqrt{\ell'\hat{\sigma}^2\ell/N}$ has uniformly asymptotically correct coverage (Corollary 3.1).

INFERENCE The proof in the appendix is short and worth following because the structure is the entire argument. Writing $\hat{J}_0 = \frac{1}{K}\sum_k \mathbb{E}_{n,k}[\psi^a(W;\hat{\eta}_{0,k})]$, $R_{N,1} = \hat{J}_0 - J_0$ and $R_{N,2} = \frac{1}{K}\sum_k \mathbb{E}_{n,k}[\psi(W;\theta_0,\hat{\eta}_{0,k})] - \frac{1}{N}\sum_i \psi(W_i;\theta_0,\eta_0)$, the linearity of the score gives the exact identity

$$\sqrt{N}(\tilde{\theta}_0 - \theta_0) = -(J_0 + R_{N,1})^{-1}\Big(\frac{1}{\sqrt{N}} \sum_{i=1}^N \psi(W_i;\theta_0,\eta_0) + \sqrt{N}R_{N,2} \Big). \tag{16}$$

The matrix perturbation is controlled by $(J_0 + R_{N,1})^{-1} - J_0^{-1} = -(J_0 + R_{N,1})^{-1}R_{N,1}J_0^{-1}$, and $\|R_{N,1}\| = O_P(N^{-1/2} + r_N)$ by splitting $\mathbb{E}_{n,k}[\psi^a(W;\hat{\eta})] - \mathrm{E}[\psi^a(W;\eta_0)]$ into a conditional-variance term bounded by

c_1^2/n and a bias term bounded by r_N. The term $R_{N,2}$ is where orthogonality is used: it is split into an empirical process piece, bounded in conditional mean square by $(r'_N)^2$ because $\hat{\eta}_{0,k}$ is non-random given I_k^c, and a bias piece $\sqrt{n}\|\mathrm{E}[\psi(W;\theta_0,\hat{\eta}_{0,k}) \mid I_k^c] - \mathrm{E}\psi(W;\theta_0,\eta_0)\|$, which a second-order Taylor expansion in r bounds by $\lambda_N + \lambda'_N$. Hence $\|R_{N,2}\| = O_P(N^{-1/2}r'_N + \lambda_N + \lambda'_N)$, and the Lindeberg-Feller central limit theorem and the Cramér–Wold device finish the proof. Conditioning on the auxiliary sample (Lemma 6.7) is the technical device that turns conditional convergence into unconditional convergence.

3.6 Specialisations: the four models

PAPER The paper verifies Assumptions 3.1 and 3.2 in four models. In each case the nuisance rate condition is a product condition.

Table 1. **Scores, nuisance parameters and rate conditions in the four specialisations**

Model and target	Orthogonal score	Product rate condition
Partially linear regression, θ_0 in $Y = D\theta_0 + g_0(X) + U$	$(Y - D\theta - g(X))(D - m(X))$, nuisance (g, m)	$\|\hat{m}_0 - m_0\|_{P,2}\,\|\hat{g}_0 - g_0\|_{P,2} \le \delta_N N^{-1/2}$
Partially linear IV, instrument $Z = m_0(X) + V$	$(Y - D\theta - g(X))(Z - m(X))$, nuisance (g, m)	Same product, with $\hat{g}_0$ and $\hat{m}_0$
ATE, binary D, $\theta_0 = \mathrm{E}[g_0(1,X) - g_0(0,X)]$	Augmented inverse propensity weighted score, Equation (17)	$\|\hat{m}_0 - m_0\|_{P,2}\,\|\hat{g}_0 - g_0\|_{P,2} \le \delta_N N^{-1/2}$, with overlap $\varepsilon \le m_0 \le 1 - \varepsilon$
LATE, binary D and binary Z	Ratio of two orthogonal reduced-form scores, nuisance (μ, m, p)	$\|\hat{p}_0 - p_0\|_{P,2}(\|\hat{\mu}_0 - \mu_0\|_{P,2} + \|\hat{m}_0 - m_0\|_{P,2}) \le \delta_N N^{-1/2}$

Source: reproduced from Sections 4 and 5 of the paper (official). The wording of the column headings is the reviewer's.

PAPER For the average treatment effect with $D \in \{0,1\}$, $Y = g_0(D,X) + U$ and propensity score $m_0(X) = \mathrm{P}(D = 1 \mid X)$, the score of Robins and Rotnitzky (1995) is

$$\psi(W;\theta,\eta) = g(1,X) - g(0,X) + \frac{D\,(Y - g(1,X))}{m(X)} - \frac{(1-D)\,(Y - g(0,X))}{1 - m(X)} - \theta. \tag{17}$$

INFERENCE Orthogonality can again be verified in a few lines. Conditioning on X, $\mathrm{E}[D(Y - g(1,X)) \mid X] = m_0(X)\,\delta_1(X)$ with $\delta_1 = g_0(1,\cdot) - g(1,\cdot)$, and similarly $\mathrm{E}[(1-D)(Y - g(0,X)) \mid X] = (1 - m_0(X))\,\delta_0(X)$. Collecting terms,

$$\mathrm{E}\big[\psi \mid X\big] = \big[\tau_0(X) - \theta\big] + \big(m_0(X) - m(X)\big)\Big[\frac{\delta_1(X)}{m(X)} + \frac{\delta_0(X)}{1 - m(X)}\Big], \tag{18}$$

where $\tau_0(X) = g_0(1,X) - g_0(0,X)$. The first bracket integrates to zero at θ_0. The second term is the product of the propensity error and the outcome-regression errors, so the AIPW score is doubly robust and Neyman orthogonal for the same reason as in the linear case. The $1/m$ and $1/(1-m)$ factors also show why overlap is a maintained assumption: the product term is amplified where the propensity is close to zero or one.

PAPER The efficiency statements are as follows. The score in Equation (17) and its ATTE analogue are efficient, so the estimators reach the bound of Hahn (1998). In the partially linear model, $\sigma^2 = (\mathrm{E}V^2)^{-1}\mathrm{E}[V^2U^2](\mathrm{E}V^2)^{-1}$, which reduces to $\mathrm{E}[U^2]/\mathrm{E}[V^2]$ under conditional homoskedasticity and then attains the semiparametric bound.

PAPER Remarks 4.10 and 5.12 give the sparse benchmark. If g_0 and m_0 are sparse with indices $s^g, s^m \ll N$ and estimated by ℓ_1 methods at rates $\sqrt{s^g/N}$ and $\sqrt{s^m/N}$, the product condition reads

$$\sqrt{s^g/N}\sqrt{s^m/N} \ll N^{-1/2} \iff s^g s^m \ll N, \tag{19}$$

which is much weaker than the requirement $(s^g)^2 + (s^m)^2 \ll N$ without sample splitting. For example, a very sparse propensity function permits a regression function with $s^g > \sqrt{N}$.

4 Data and Empirical Strategy

4.1 Data

PAPER The paper uses three real data sets and two simulations. The text read for this review does not report the number of observations for the first two applications, so those cells are marked as not stated.

Table 2. **Data used in the paper's applications**

Application	Outcome Y and treatment D	Controls X	Sample
Pennsylvania Reemployment Bonus experiment (US Department of Labor, 1980s)	Y: log duration of unemployment. D: assignment to the most generous bonus scheme (treatment 4), other schemes dropped	Age group dummies, gender, race, dependents, quarter, location, recall expectation, occupation type; 96 terms for lasso	Randomised; number of observations not stated in the text read
401(k) eligibility, 1991 Survey of Income and Program Participation, as in Chernozhukov and Hansen (2004)	Y: net financial assets. D: 401(k) eligibility, or participation instrumented by eligibility	Age, income, family size, education, married, two-earner, defined benefit pension, IRA participation, home ownership	Cross-section of households; number of observations not stated in the text read
Institutions and growth, Acemoglu, Johnson and Robinson (2001)	Y: log GDP per capita. D: protection against expropriation risk. Z: early settler mortality	Distance from the equator and four continent dummies; second-order terms for lasso	64 countries

Source: Section 6 of the paper (official). Numbers of observations are reported as not stated rather than inferred.

PAPER Nuisance functions are fitted by five single learners, a single CART tree with penalty chosen by ten-fold cross-validation, a random forest averaging 1000 trees, boosted regression trees with cross-validated tuning, a neural network with two neurons and decay 0.02 (eight neurons and decay 0.01 in the 401(k) application), and lasso with the Belloni et al. (2012) data-driven penalty.

Two hybrids are added: an ensemble whose convex weights minimise five-fold cross-validated squared prediction error, and a ”best” learner chosen separately for each nuisance function. The ensemble drops the neural network in the 64-country application because of instability. In the 401(k) application propensity scores are trimmed at 0.01 and 0.99.

4.2 Empirical specification and identification

PAPER The estimating equations are Equation (4) for the partially linear model, Equation (17) for the average treatment effect in the interactive model, the LATE score for 401(k) participation instrumented by eligibility, and the partially linear IV score for institutions instrumented by settler mortality. Each estimate is DML2, computed over 100 random partitions, with the median estimate and two standard errors reported: the median across splits of the within-split standard error, in brackets, and the standard error that adds across-split dispersion through Equation (13), in parentheses. In the bonus experiment the propensity score is the fraction treated, because assignment was random.

CRITIQUE The identifying variation has two quite different sources, and the review keeps them separate. In the bonus experiment it is random assignment, so θ_0 is causal by design and the method serves only to increase precision and to allow heterogeneity. In the 401(k) application it is the maintained assumption of Poterba, Venti and Wise (1994) that eligibility is exogenous conditional on income and job-related characteristics. The machine learner does not weaken that assumption. It relaxes only the functional form by which income and the other controls enter. In the 64-country application identification comes from settler mortality as an instrument, whose exclusion restriction is argued by Acemoglu et al. (2001) and not tested here. The paper is explicit that relaxing the linear control for geography is the only change.

4.3 Data limitations and biases

- *Unobserved confounding.* Orthogonality removes bias from nuisance estimation error, not from omitted variables. In the 401(k) data a household with a taste for saving selects into firms offering a plan whatever the control set. No sensitivity analysis is reported.
- *Small samples relative to the asymptotics.* The 64-country application has $N = 64$, with each fold in a five-fold split holding about 13 countries and the learner trained on about 51. A rate condition of the form $o(N^{-1/4})$ is not checkable at that size, and the neural network is dropped from the ensemble for instability.
- *Learner tuning.* Tuning parameters are chosen by cross-validation on the same data used downstream, with no nested validation. This leaves the theory intact only if the tuned learner satisfies the rate condition.
- *Weak instruments.* For LATE and for partially linear IV the denominator $\mathrm{E}[m(1,X)-m(0,X)]$ or $\mathrm{E}[DV]$ must be bounded away from zero (Assumptions 5.8d and 4.6c). With a weak first stage the normal approximation is poor and the paper offers no diagnostic.
- *Selection of examples.* Three favourable applications cannot reveal how often the method fails. This is a mild form of publication bias in the empirical section.

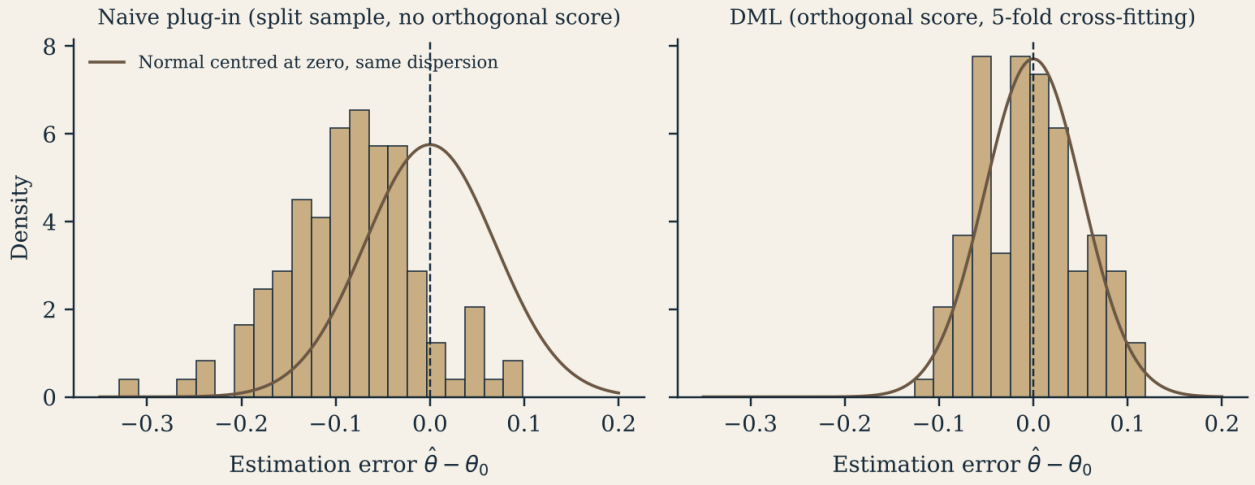
5 Results

5.1 Simulation evidence in the paper

EVIDENCE Figure 1 of the paper draws the sampling distribution of the naive estimator in Equation (2) when g_0 is learned by a random forest in an experiment the authors describe as favourable to forests (g_0 smooth in a few variables). The histogram is displaced to the right and visibly different from the normal curve that ignores bias, whereas the DML histogram, built from identical simulated data, is centred and close to the normal approximation. Figure 2 isolates overfitting with the contrived learner in Equation (11): the full-sample estimator is shifted markedly left, and the two-fold cross-fitted estimator is centred with roughly the same spread. The paper does not report the sample size, the number of replications or the numerical bias in the text read, so the figures can be read qualitatively only.

EVIDENCE To check the qualitative message with numbers, the reviewer ran a small simulation of the same kind. The design is generated by the reviewer and is not a replication of the authors' code: $N = 400$, $p = 20$ standard normal controls, $m_0(X) = \sin X_1 + 0.5 \cos 2X_2 + 0.25 X_3$, $g_0(X) = \Lambda(X_1) + 0.5 \sin X_2 + 0.25 X_3$ with Λ the logistic function, $D = m_0(X) + V$, $Y = 0.5D + g_0(X) + U$ with standard normal U and V, and random forests of 30 trees with leaves of at least 5 observations. The naive estimator learns the regression of Y on X on a half sample and applies Equation (2) on the other half. The DML estimator uses the partialling-out score and five-fold cross-fitting. Both are computed over 120 replications.

Figure 1. **Sampling distribution of the estimation error, naive plug-in versus DML (120 replications, $N = 400$, $\theta_0 = 0.5$)**



Source: generated by the reviewer (not from the paper). Left: mean error -0.085, standard deviation 0.069. Right: mean error -0.004, standard deviation 0.052. Random forests with 30 trees, $p = 20$ controls. The curve is a normal density centred at zero with the same standard deviation as the histogram.

EVIDENCE The naive estimator carries a mean error of -0.085, which is 17 per cent of $\theta_0 = 0.5$ and about 1.2 of its own standard deviations (0.069), while the DML mean error is -0.004, within one Monte Carlo standard error ($0.052/\sqrt{120} \approx 0.005$) of zero. The naive estimator is biased downward here because the learner of $\mathrm{E}[Y \mid X]$ absorbs the part $\theta_0 m_0(X)$ of the outcome that is explained by X through the treatment, which is exactly the regularisation-bias channel in Equation (3). The DML

standard deviation is also smaller, since partialling out removes variation in D that is explained by X. The caveat is that this is one data generating process with a small sample.

5.2 Pennsylvania bonus experiment

EVIDENCE Table 3 (first block) reports the average treatment effect of the most generous bonus on log unemployment duration. Every learner returns a negative estimate between -0.073 and -0.085 with standard errors near 0.036, so every t statistic is about -2. In the interactive model with five folds the best-learner estimate is -0.077, with $t = -0.077/0.036 \approx -2.1$. Because the outcome is logged, $e^{-0.077} - 1 \approx -7.4$ per cent: the bonus shortens the expected unemployment spell by about seven per cent, with a 95 per cent interval running roughly from -14 to -1 per cent. The effect is statistically significant at the five per cent level in all specifications but economically modest and imprecise. The partially linear and interactive panels agree closely (-0.074 against -0.077), which is expected under randomisation and carries no information about the validity of the method beyond precision.

5.3 401(k) eligibility and participation

EVIDENCE Without controls, eligibility is associated with \$19,559 more net financial assets (standard error 1,413). With flexible controls and five folds the interactive-model estimates fall to between \$7,170 (lasso) and \$8,105 (forest), and the best-learner estimate is \$7,753, so the unconditional gap shrinks by about 60 per cent ($1 - 7{,}753/19{,}559 \approx 0.60$). The partially linear estimates are higher, \$8,187 to \$9,247, with a best estimate of \$9,215, still 53 per cent below the uncontrolled gap. The conclusion is that the large majority of the raw correlation reflects the observed characteristics that drive both eligibility and wealth, and that the remaining effect is a few thousand dollars, significant at standard levels ($7{,}753/1{,}294 \approx 6.0$ using the split-adjusted standard error). For participation instrumented by eligibility the LATE estimates sit at \$8,944 (lasso) and \$11,113 to \$11,764 for the others, to be compared with the linear IV benchmark of \$13,102 (standard error 1,922). The flexible estimates are thus about 15 per cent smaller than the linear IV estimate, which the authors read as evidence that the Poterba et al. control specification is somewhat too simple.

5.4 Institutions and economic output

EVIDENCE The coefficient on property-rights protection in a partially linear IV model ranges from 0.73 (boosting, five folds) to 1.00 (neural network, five folds), with a best-learner estimate of 0.88 and the two reported standard errors of 0.41 (median within split, in brackets) and 0.39 (in parentheses), so t lies between 2.1 and 2.3. Acemoglu et al. report 1.10 with standard error 0.46. Interpreting the coefficient as a semi-elasticity, one additional unit of the institutions index is associated with roughly a 0.8 to 0.9 log-point (about 130 to 140 per cent) higher output per head, a number that is only meaningful as a comparison with the original study. The DML estimates are about 20 per cent smaller but statistically indistinguishable from the original, because the sampling uncertainty is of the same order as the shift.

Table 3. **DML estimates in the three applications, five-fold cross-fitting, median over 100 splits**

Specification	Lasso	Tree	Forest	Boost	Net	Ensemble	Best
Bonus, interactive ATE (log duration)	−0.081	−0.085	−0.074	−0.077	−0.073	−0.078	−0.077
split-adjusted standard error	0.036	0.037	0.036	0.036	0.036	0.036	0.036
401(k) eligibility, interactive ATE (USD)	7,170	7,993	8,105	7,713	7,788	7,839	7,753
split-adjusted standard error	1,398	1,236	1,299	1,177	1,293	1,148	1,294
401(k) eligibility, partially linear (USD)	8,187	8,871	9,247	9,110	9,038	9,166	9,215
split-adjusted standard error	1,558	1,418	1,328	1,328	1,355	1,310	1,312
401(k) participation, LATE (USD)	8,944	11,459	11,764	11,133	11,186	11,173	11,113
split-adjusted standard error	3,307	1,786	1,893	1,710	1,890	1,678	1,675
Institutions, partially linear IV (log GDP)	0.77	0.95	0.90	0.73	1.00	0.83	0.88
standard error, median within split	0.24	0.46	0.41	0.33	0.33	0.37	0.41
standard error, in parentheses	0.17	0.45	0.40	0.27	0.30	0.34	0.39

Source: Tables 1 to 4 of the paper (official), five-fold rows. Unless stated, standard errors are those adjusted for variation across sample splits by the median method; in the institutions block both reported versions are shown. The reviewer's percentage and t statistic calculations in the text are computed from these numbers.

5.5 What Table 3 answers and fails to answer

EVIDENCE The table answers one question: does the choice of learner change the point estimate? The dependent variable differs by block, the key regressor is the treatment, and columns change only the learner used to estimate nuisance functions. The most important number is the dispersion across columns relative to the standard error: in the 401(k) interactive block the spread of 935 (from 7,170 to 8,105) is about 0.7 standard errors, and in the bonus block the spread is 0.012, a third of a standard error. The coefficient implies that the estimates are practically learner-invariant in these examples, as the theory predicts if every learner satisfies the product-rate condition. The exceptions are informative. Lasso in the LATE block is 2,170 to 2,820 below the other learners and has a standard error nearly double the others (3,307 against 1,675 to 1,893), consistent with the authors' explanation that a linear learner extrapolates badly when the main fold has covariate values outside the range of the auxiliary folds. The gap between the interactive and the partially linear 401(k) estimates, $1,462 at the best learner, is 1.1 standard errors and exceeds the spread across learners within either model (935 and 1,060), which indicates that the functional form of the treatment effect matters more than the learner. The table does not establish that the estimate is correct, since all columns share the same identification assumption and the same data, and it provides no coverage evidence.

6 Robustness and Additional Tests

Table 4. Robustness tests in the paper: concern, test, result and assessment

Concern	Test reported	Result	Does it resolve the concern?
Estimate depends on one learner	Seven learners, including ensemble and best-of selection	Estimates agree within about one standard error in all applications; lasso is the outlier for LATE	Partly. Agreement rules out a learner-specific artefact, but all learners may share the same bias if the nuisance is poorly approximated by all of them
Random partition drives the result	100 splits, median estimate, split-adjusted standard errors	Standard errors rise most for lasso (16 to 46 per cent in the 401(k) blocks) and by a few per cent for the other learners	Yes for partition noise. Corollary 3.3 covers fixed S only; S growing with N is left open
Number of folds	Two versus five folds	Five-fold standard errors are smaller in 401(k), larger in the institutions example	Partly. No theory predicts the direction; the authors flag this as an open question
Functional form of the effect	Interactive versus partially linear model in 401(k)	Best estimates $7,753 and $9,215	No. A gap of 1.1 standard errors is left unexplained, and it signals that a constant coefficient may be misspecified
Extreme propensity weights	Trimming at 0.01 and 0.99 in the 401(k) application	Effect of trimming not reported	No. Overlap sensitivity is untested
Propensity estimation in an experiment	Known fraction treated versus ML propensity	Results similar and unreported (footnote)	Yes for randomised treatment, uninformative for observational data
Bias of naive estimator in practice	Figure 1 simulation with random forest	Naive estimator displaced, DML centred	Yes in the designed experiment; one data generating process, size unreported
Overfitting	Figure 2 contrived learner, with and without splitting	Full-sample estimator displaced, cross-fit centred	Illustrative only, since the learner is artificial
Unconfoundedness	Comparison of controlled estimate with the uncontrolled gap	Estimate shrinks by about 60 per cent	No. Attenuation shows controls matter, not that they suffice

Source: Sections 1 and 6 of the paper (official) for the tests and results; the final column is the reviewer's assessment.

CRITIQUE The robustness battery is orthogonal to the main threats. The tests address nuisance-estimation noise very well and identification not at all. The central assumption, selection on observables with overlap, is never stressed, and neither is the rate condition: the paper does not report out-of-sample prediction errors of the individual nuisance functions, which would be the natural proxy for $\|\hat{m}_0 - m_0\| \cdot \|\hat{g}_0 - g_0\|$. A reader therefore cannot judge whether the product condition is plausible in each application. A direct check would report cross-fitted root mean squared errors for $\hat{m}_0$ and $\hat{g}_0$ and compare their product with $N^{-1/2}$.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Under Assumptions 3.1 and 3.2, the DML estimators are asymptotically normal and the variance estimator is consistent, uniformly over the stated class of distributions. This is a mathematical result with a transparent proof, and the review reproduced its core steps. The exact product structure of the bias for the partially linear and AIPW scores (Equations 6 and 18) is also established, and it shows precisely why the cross-term of the nuisance errors is the only first-order channel left. Naive plug-in estimation with a non-orthogonal score is biased at a rate slower than $N^{-1/2}$ when the learner converges slower than $N^{-1/2}$.

SUPPORTED BUT CONDITIONAL

That the estimator is insensitive to the choice of learner. This follows from the theory only if every learner satisfies the product-rate condition, and in the applications it is observed, not tested. The recommendation of DML2 over DML1 and of five folds over two is supported by the authors' experience and a heuristic about Jacobian stability, not by a theorem. The causal reading of the 401(k) and institutions estimates depends on exogeneity assumptions taken from earlier studies.

SUGGESTIVE

That the Poterba, Venti and Wise control specification is ”somewhat too simplistic”, inferred from a 15 per cent difference between the flexible LATE and the linear IV estimate, which is well inside one standard error (1,922) of the linear estimate. That lasso's larger split-adjusted standard error reflects extrapolation, offered as a belief by the authors, without a test.

NOT ESTABLISHED

That DML delivers correct finite-sample coverage in empirical settings of the size analysed; no coverage simulation is reported. That the rate condition holds for the learners actually used, since no prediction errors are shown. That the method protects against unobserved confounding, which it cannot by construction. That the number of folds or the number of repeated splits has a systematic effect on precision; the paper reports contradictory patterns and leaves this open.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Generality: a single high-level theorem covers any learner that meets the rate condition, so progress in machine learning rates transfers automatically to inference. This matters because the econometrician need not re-derive theory for each new learner.
- Weak conditions: no Donsker or entropy growth restriction, and a product-rate requirement much weaker than the sparsity requirements without splitting (Equation 19).
- Uniformity over $P \in \mathcal{P}_N$, which protects against the failure of pointwise asymptotics near the boundary of the parameter space.
- Practicality: the procedure is implementable with off-the-shelf learners and produces a valid standard error.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The constructive taxonomy of orthogonal scores in Section 2 is itself a lasting contribution. It turns ”find a doubly robust or locally robust score” into a procedure for any moment model, and it unifies the AIPW, Robinson, Neyman and influence-function approaches.
- The exact identity in Equation (6) makes the method easy to teach and to audit: the bias is the covariance of two errors, so the analyst can see which nuisance must be estimated well.
- Reporting both within-split and split-adjusted standard errors is an honest treatment of the extra uncertainty introduced by the random partition, which most applied work ignores.
- The empirical illustrations include a randomised benchmark, which validates the approach where the truth is approximately known.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The theory delivers inference conditional on a rate condition on the product of two learned functions, but nothing in the method or in the empirical sections measures or tests that condition. **Why it matters.** The result is a statement of the form ”if the learners are good enough, then coverage is correct”, and in finite samples with p comparable to N the condition may fail without any visible symptom, since the estimate would still be stable across learners that share a common failure. **Potential bias.** Residual regularisation bias of order $\sqrt{N}\|\hat{m}_0 - m_0\| \, \|\hat{g}_0 - g_0\|$ would shift the estimate relative to its standard error. With each error of order $N^{-0.3}$ and $N = 10{,}000$ the bias term in Equation (5) is roughly $N^{0.5-0.6} = N^{-0.1} \approx 0.40$ standard errors, and with $N^{-0.2}$ errors it is $N^{0.1} \approx 2.5$ standard errors. **How it could be tested.** Report cross-fitted root mean squared errors for each nuisance function, compare their product with $N^{-1/2}$, and add a Monte Carlo calibration in which the learner is applied to data simulated from the fitted nuisance functions with a known θ_0.

OTHER WEAKNESSES

- *Unconfoundedness and overlap are maintained, not tested.* Problem: orthogonality is silent about omitted variables. Why it matters: the 401(k) estimate remains a selection-on-observables estimate. Potential bias: unobserved saving preference biases the effect upward. Test: sensitivity bounds on the omitted-variable bias in the Riesz representer formulation.
- *Instability of the AIPW weights.* Problem: terms $1/m$ and $1/(1-m)$ with trimming at 0.01 means weights up to 100. Why it matters: a few observations dominate the variance and the normal approximation is poor. Potential bias: finite-sample bias and wrong standard errors. Test: report the distribution of cross-fitted propensity scores and the effective sample size.
- *Hyperparameter tuning is not accounted for.* Problem: learners are tuned by cross-validation on the full sample before cross-fitting. Why it matters: tuning uses outcome information from the evaluation folds. Potential bias: a mild form of overfitting not covered by Assumption 3.2. Test: nested cross-fitting with tuning inside each training fold.
- *Median aggregation across splits has no distributional theory beyond fixed S.* Problem: the recommended median and median-variance methods are justified only by first-order equivalence for fixed S. Why it matters: the variance adjustment may be conservative or liberal. Test: coverage simulation with S growing.
- *Independent sampling.* Problem: i.i.d. data are assumed throughout. Why it matters: panel, clustered and time-series data, which dominate economics and finance, violate random fold assignment. Potential bias: leakage across folds, so the training and evaluation samples are dependent and the key conditioning argument fails. Test: block or purged cross-fitting with simulation.
- *Moment conditions on the data.* Problem: $\|Y\|_{P,q} \leq C$ with $q > 4$ is required in Assumptions 4.5 and 4.6. Why it matters: outcomes with infinite fourth moments are excluded. Test: tail diagnostics of the residuals.

8 Contradictions and Edge Cases

CRITIQUE The following exercises try to break the result.

CRITIQUE *When the contrived overfitting bias vanishes.* In the contrived example $\hat{g}_0(X_i) - g_0(X_i) = (Y_i - g_0(X_i))/N^{1/2-\kappa} = (D_i\theta_0 + U_i)/N^{1/2-\kappa}$. Multiplying by V_i, summing, and dividing by $\sqrt{N}$ gives $N^{\kappa}\theta_0 \,\mathrm{E}[V^2] + O_P(N^{-1/2+\kappa})$, because $\mathrm{E}[VD] = \mathrm{E}[V^2]$ and $\mathrm{E}[VU] = 0$. The divergence therefore scales with θ_0 and disappears when the true effect is zero. With $\theta_0 = 0.5$, $\mathrm{E}[V^2] = 1$, $\kappa = 0.1$ and $N = 10^4$ the bias is $0.5 \times 10^{0.4} \approx 1.26$ in units of the asymptotic standard deviation $\sigma = 1$ (reviewer's calculation), which is large enough to produce the leftward shift in the paper's Figure 2 and also shows that a test of the null $\theta_0 = 0$ would be correctly sized even without splitting in this example while a test of any nonzero value would not.

CRITIQUE *Very strong confounding.* If $\mathrm{Var}(V) = \mathrm{E}[V^2]$ is small relative to $\mathrm{Var}(D)$, so that X almost determines D, the denominator $\hat{J}_0 = \frac{1}{K}\sum \mathbb{E}_{n,k}[\hat{V}_i D_i]$ is small and the asymptotic variance

$\mathrm{E}[V^2U^2]/(\mathrm{E}V^2)^2$ explodes. Example: if $\mathrm{E}[V^2] = 0.05$ and $\mathrm{E}[U^2] = 1$, then $\sigma^2 = \mathrm{E}[U^2]/\mathrm{E}[V^2] = 20$ under homoskedasticity, a standard deviation of 4.5 times that of the case with $\mathrm{E}[V^2] = 1$. Assumption 4.5 requires $\mathrm{E}V^2 \geq c$, but gives no guidance about how small c can be before the normal approximation is unusable. The same issue is known in ATE estimation as the overlap problem.

CRITIQUE *Propensity near the boundary.* In the ATE variance $\sigma^2 = \mathrm{E}[\psi^2]$ the contribution of the treated group is of order $\mathrm{E}[\sigma_1^2(X)/m_0(X)]$. For a propensity uniformly distributed on $(\varepsilon, 1-\varepsilon)$ and unit conditional variance, $\mathrm{E}[1/m_0] = \ln((1-\varepsilon)/\varepsilon)/(1-2\varepsilon)$, which equals about 4.7 at $\varepsilon = 0.01$ and 6.9 at $\varepsilon = 0.001$ (reviewer's calculation). The growth is slow for a uniform density, but any propensity distribution with positive mass density at zero produces a much faster blow-up, and the trimming used in the 401(k) example changes the estimand because it conditions on the region of overlap.

CRITIQUE *When the rate condition fails in the other direction.* The product bound $\sqrt{n}\, n^{-(\varphi_m+\varphi_g)}$ needs $\varphi_m + \varphi_g > 1/2$. A learner that is only consistent ($\varphi \to 0$) paired with a well-estimated nuisance is enough: this is the double robustness. But if both nuisances live in a nonparametric class of smoothness s in dimension d, minimax rates are $n^{-s/(2s+d)}$ and the requirement becomes $2s/(2s+d) > 1/2$, that is $s > d/2$. With $d = 10$ controls, each nuisance needs more than five derivatives. DML therefore does not escape the curse of dimensionality, it converts it into a smoothness-versus-dimension condition on the nuisances, and sparse structure or low intrinsic dimension must come from somewhere else.

CRITIQUE *Alternative mechanism.* The approximate agreement of learners in the 401(k) example, together with the closeness to the original Poterba et al. estimates, is equally consistent with a world in which the true nuisance functions are nearly linear and all learners approximate them well, and with a world in which no available covariate is informative about the unobserved confounder. The data cannot distinguish these cases.

CRITIQUE *Samples of 64 observations.* With $N = 64$ and $K = 5$ the evaluation folds contain about 13 countries and each nuisance learner is trained on about 51. A reading of Table 4 also exposes an inconsistency. The text says that adjusting for variation across splits increases the standard errors, and the median-variance formula in Equation (13) cannot do otherwise, because it adds a non-negative dispersion term to each split's variance. Yet in every column of Table 4 the number in parentheses is smaller than the number in brackets (for example 0.17 against 0.24 for lasso with five folds), whereas in Tables 1 to 3 the parenthesised figure is the larger one. Either the two rows are transposed in the version read or the table reports something other than the caption states; the review cannot resolve this from the text and therefore reports both figures. Asymptotic normality at $N = 64$ is in any case an act of faith, and the authors do not claim otherwise.

9 Further Literature

Table 5. Literature connections

Paper	Shared idea	Relationship	Why it matters
Neyman (1959)	Orthogonal score for composite hypotheses, the $C(\alpha)$ statistic	Preceded and supplied the foundation	Defines the construction that Section 2.2.1 generalises
Robinson (1988)	Partialling out in the partially linear model with root-n nonparametric estimation	Preceded; the lead model is Robinson's	DML replaces kernel nuisance estimation with arbitrary learners
Andrews (1994) and Newey (1994)	Stochastic equicontinuity and the influence function of a semiparametric estimator	Preceded; foundation of the orthogonality and influence-function recipes	They require Donsker-type conditions that DML bypasses with cross-fitting
Robins and Rotnitzky (1995) and Hahn (1998)	Augmented inverse propensity weighting and the efficiency bound for the ATE	Preceded; supply the ATE score and the bound	Shows the ATE results are an application of the general framework
Belloni, Chen, Chernozhukov and Hansen (2012); Belloni, Chernozhukov and Hansen (2014)	Post-selection inference with lasso, sample splitting for optimal instruments, double selection	Developed similar ideas under sparsity	DML extends the idea to non-sparse nuisances and non-lasso learners
van der Laan and Rubin (2006)	Targeted maximum likelihood with super learner nuisance estimation	Developed similar ideas through a different methodology	Reaches doubly robust estimation through an update step; shares sample splitting
Kang and Schafer (2007)	Performance of doubly robust estimators under misspecification and extreme weights	Contradicts the optimism about robustness in finite samples	Shows that doubly robust estimators can behave worse than simple alternatives when weights are extreme
Athey, Imbens and Wager (2018)	Approximate residual balancing for ATE with high-dimensional controls	Alternative identification of the same estimand with weaker propensity conditions	Allows inconsistent propensity estimation under strong regression sparsity
Wager and Athey (2018)	Causal forests for heterogeneous effects	Applied a related cross-fitting idea to effect heterogeneity	Extends the target from an average to a conditional effect
Chernozhukov, Newey and Singh (2022)	Automatic debiased learning of the Riesz representer	Extended the contribution	Removes the need to derive the orthogonal score by hand
Kennedy (2023)	Optimal doubly robust estimation of heterogeneous effects	Improved the rate analysis for the conditional case	Shows second-order remainders also control the conditional effect
Feng, Giglio and Xiu (2020)	Double selection to test factors in the cross-section of returns	Applied the approach to asset pricing	Shows the method already reaches financial econometrics

Source: Papers cited in the text of the review. Neyman, Robinson, Andrews, Newey, Robins and Rotnitzky, Hahn, the Belloni papers, van der Laan and Rubin and Athey, Imbens and Wager are cited by the authors; Kang and Schafer, Wager and Athey, Chernozhukov, Newey and Singh, Kennedy, and Feng, Giglio and Xiu are supplied by the reviewer from the wider literature and were not read for this review.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** No diagnostic for the product-rate condition. *Why it matters:* The central assumption is invisible to the user. *Existing limitation:* Prediction errors of nuisance fits are not reported. *Way forward:* Report cross-fitted prediction errors and a Monte Carlo calibration based on the fitted nuisances.
- **Gap:** No treatment of unobserved confounding. *Why it matters:* The 401(k) and institutions results are causal only under strong assumptions. *Existing limitation:* Orthogonality is local to nuisance error. *Way forward:* Add sensitivity analysis of the estimate to the strength of omitted confounders.
- **Gap:** The effect of K and S on finite-sample precision is left as an open question. *Why it matters:* Practitioners must choose them. *Existing limitation:* Contradictory empirical patterns. *Way forward:* Simulation across designs with a theory of the variance of the aggregated estimator.

10.2 Literature-level gaps

- **Gap:** Valid cross-fitting under dependence (panels, clustering, time series). *Why it matters:* Most financial data are dependent. *Existing limitation:* Random fold assignment violates conditional independence. *Way forward:* Blocked, purged or hv-block cross-fitting with proved remainder bounds.
- **Gap:** Inference when moments are heavy-tailed. *Why it matters:* Financial returns have tail exponents near three according to empirical work on index returns. *Existing limitation:* Assumptions require $q > 4$ moments. *Way forward:* Robust scores based on truncation or median of means.
- **Gap:** Joint accounting for learner tuning, model selection and inference. *Why it matters:* Practitioners tune on the same data. *Existing limitation:* Assumptions treat the learner as given. *Way forward:* Theory for nested cross-fitting.

10.3 Methodological gaps

- **Gap:** Confidence intervals that are robust to rates slower than $N^{-1/4}$. *Why it matters:* Learners in high dimension may be slower. *Existing limitation:* Higher-order influence functions need heavy machinery. *Way forward:* Higher-order orthogonal scores, as in the work cited by the authors (Robins et al. 2008 and 2017), made practical.
- **Gap:** Finite-sample coverage evidence for the specific learners used. *Why it matters:* The asymptotics are loose guides. *Existing limitation:* One simulation each. *Way forward:* Large-scale simulation benchmark across data generating processes.

11 Research Opportunities

EXTENSION The ideas below follow from the gaps above. Data sources named are public or commercial as stated; where a number is a reviewer's estimate it is labelled as such.

Idea 1. Blocked cross-fitting for the price of a stock characteristic

RESEARCH QUESTION. Is the cross-sectional risk premium attached to a candidate characteristic still significant once a high-dimensional set of other characteristics is controlled for flexibly, with inference that remains valid under cross-sectional and serial dependence?

MOTIVATION. Hundreds of published characteristics compete to explain returns, and the standard answers (Fama-MacBeth regressions, double-selection lasso) either impose linearity or ignore dependence across folds.

HYPOTHESIS. A material share of characteristics that appear priced in linear controlled regressions lose significance when the controls enter nonparametrically and the folds respect time.

ECONOMIC MECHANISM. If a characteristic proxies a nonlinear function of other characteristics, linear controls leave a residual correlation that the partially linear model with learned $g_0(X)$ removes.

LITERATURE CONNECTION. Chernozhukov et al. (2018) for the score, Feng, Giglio and Xiu (2020) for double selection, and Racine (2000) and Lopez de Prado (2018) for dependent-data cross-validation.

DATA. Open Source Cross-Sectional Asset Pricing data of Chen and Zimmermann (2022), built from CRSP and Compustat; monthly, 1963 onward.

VARIABLES. Y: excess return next month. D: standardised candidate characteristic. X: all other characteristics plus industry and size indicators.

METHOD. DML2 with the partially linear score, random forest and lasso learners, folds formed by calendar blocks with an embargo of one month, Newey-West type variance on the pooled score.

IDENTIFICATION STRATEGY. Conditional exogeneity of the characteristic given other characteristics; a causal reading is not claimed, only a conditional premium.

EXPECTED CONTRIBUTION. A re-ranking of published characteristics under nonparametric controls, plus a simulation of the size distortion of random-fold cross-fitting under dependence.

MAIN RISK. Characteristics are noisy, so the estimates may be too imprecise to discriminate, and cross-sectional correlation of residuals within a month may not be fully captured by the variance estimator.

Idea 2. Orthogonal scores under heavy-tailed outcomes

RESEARCH QUESTION. How does DML coverage deteriorate when the outcome has a tail exponent close to three, and does a truncated or median-of-means orthogonal score restore it?

MOTIVATION. Assumptions 4.5 and 4.6 require $q > 4$ moments, while empirical work on index returns reports tail exponents near three, which implies an infinite fourth moment.

HYPOTHESIS. Coverage of nominal 95 per cent intervals falls below 90 per cent at moderate N under Student-t noise with three degrees of freedom, and a robust score recovers it.

ECONOMIC MECHANISM. The variance estimator $\hat{\sigma}^2$ involves $\mathbb{E}_{n,k}[\psi^2]$, whose own sampling error is governed by a fourth moment of the residuals.

LITERATURE CONNECTION. Chernozhukov et al. (2018), Theorem 3.2 on the variance estimator, and the tail evidence in Gopikrishnan, Plerou, Amaral, Meyer and Stanley (1999).

DATA. Simulated designs with controlled tail index; then daily returns of public equity indices as the outcome in a predictive regression.

VARIABLES. Y: next-day return. D: lagged signed volume or a sentiment proxy. X: lagged returns, volatilities and calendar indicators.

METHOD. Monte Carlo of coverage and length for DML2, a Huberised score, and a median-of-means variance estimator.

IDENTIFICATION STRATEGY. Not causal; the exercise targets inference validity for a conditional association.

EXPECTED CONTRIBUTION. A rule of thumb for when to trust DML intervals in financial data and a robust variant.

MAIN RISK. Truncation changes the estimand when the score is asymmetric, so unbiasedness must be re-established.

Idea 3. Causal price impact with orthogonal scores in the limit order book

RESEARCH QUESTION. What is the effect of signed order flow on the next price change when the state of the book is controlled for nonparametrically, and how does it compare with the linear impact coefficients in the literature?

MOTIVATION. Linear impact regressions condition on a handful of book variables, while the true confounders (queue sizes, spread, recent volatility) are numerous and nonlinear.

HYPOTHESIS. The partially linear estimate is smaller than the unconditional regression coefficient and decreases with the richness of the controls.

ECONOMIC MECHANISM. Order flow and price changes share common drivers in the book state and in news, and partialling out the predictable part of flow isolates the impact of the unexpected component.

LITERATURE CONNECTION. Cont, Kukanov and Stoikov (2014) for order flow imbalance, Donier, Bonart, Mastromatteo and Bouchaud (2015) for nonlinear impact, and Chernozhukov et al. (2018) for the inference.

DATA. LOBSTER or Nasdaq ITCH message data, five-minute intervals for a sample of liquid stocks (commercial data; free sample files exist for a few tickers).

VARIABLES. Y: mid-price change over the interval. D: order flow imbalance. X: book depth at several levels, spread, lagged returns, time of day.

METHOD. Partially linear DML with blocked folds by day, and an IV version using exogenous flow shocks such as index rebalancing as instruments.

IDENTIFICATION STRATEGY. Conditional exogeneity of flow given the book is doubtful, so the IV version is the credible one, and the OLS-type version is reported as a bound.

EXPECTED CONTRIBUTION. A clean statement of how much of measured impact is attributable to confounding by book state.

MAIN RISK. Instruments tied to scheduled events are rare and may be weak; high-frequency microstructure noise inflates variance.

Idea 4. Sensitivity and replication of the 401(k) estimate with current survey data

RESEARCH QUESTION. How large would unobserved confounding have to be to erase the DML estimate of 401(k) eligibility, and does the estimate replicate in a more recent cross-section?

MOTIVATION. The 60 per cent attenuation from the unconditional gap shows controls matter, but the paper offers no bound on what controls could not capture.

HYPOTHESIS. Confounding of the strength observed for income would suffice to reduce the effect by half, and the replication yields a smaller effect than 1991.

ECONOMIC MECHANISM. Employers offering plans differ in unobserved ways correlated with saving propensity; later cohorts face automatic enrolment, which changes the participation margin.

LITERATURE CONNECTION. Poterba, Venti and Wise (1994, 1995), Chernozhukov et al. (2018), Chernozhukov, Cinelli, Newey, Sharma and Syrgkanis (2022) for bounds based on the Riesz representer, and Oster (2019).

DATA. 1991 SIPP extract distributed with the *hdm* package for the replication; Survey of Consumer Finances 2019 and 2022 from the Federal Reserve for the update.

VARIABLES. Y: net financial assets. D: eligibility. X: demographic and financial controls as in the paper.

METHOD. DML for the ATE with the AIPW score, then sensitivity parameters for the share of residual variance in the outcome and in the Riesz representer explained by an omitted factor.

IDENTIFICATION STRATEGY. Selection on observables with a bounded departure.

EXPECTED CONTRIBUTION. Quantified robustness for a canonical application and a test of external validity across decades.

MAIN RISK. The SCF differs in definitions of wealth and in sampling design, so the two estimates may not be directly comparable.

Idea 5. Semi-synthetic calibration of the product-rate condition

RESEARCH QUESTION. Can coverage of DML intervals be estimated for a given data set by simulating from fitted nuisance functions, and does the calibrated coverage predict the true coverage?

MOTIVATION. The weakest link, the nuisance rate condition, is not observable, and a direct test is impossible, but a Monte Carlo built from the fitted nuisance functions with a known θ_0 gives an empirical proxy.

HYPOTHESIS. Calibrated coverage falls below nominal exactly in the applications where the product of cross-fitted nuisance errors is large relative to $N^{-1/2}$.

ECONOMIC MECHANISM. None beyond the statistical one: the bias term in Equation (5) is a deterministic function of nuisance errors.

LITERATURE CONNECTION. Chernozhukov et al. (2018) for the theory, Kang and Schafer (2007) for the finite-sample behaviour of doubly robust estimators.

DATA. The three public applications in the paper, using the replication data in the *hdm* package.

VARIABLES. As in the original applications.

METHOD. Fit nuisances, simulate outcome and treatment from them with added bootstrap residuals, re-estimate with each learner and fold number, and record bias, coverage and interval length.

IDENTIFICATION STRATEGY. Not applicable; the target is the performance of the estimator in a world where θ_0 is known.

EXPECTED CONTRIBUTION. A diagnostic report that complements every DML estimate with an estimated coverage.

MAIN RISK. Simulated data inherit the fitted model and may understate difficulty, so the calibration may be optimistic.

11.1 Ranking

Table 6. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Blocked cross-fitting, characteristics	3	4	5	4	5	4	25
2. Heavy-tailed orthogonal scores	5	3	4	4	4	5	25
5. Semi-synthetic calibration	4	4	5	4	3	4	24
4. 401(k) sensitivity and replication	2	5	5	3	3	3	21
3. Price impact in the order book	4	2	3	3	4	4	20

Source: reviewer's subjective scoring; the table is sorted by total, and ties are broken by feasibility.

The top three ideas are the blocked cross-fitting study of characteristics, the heavy-tailed orthogonal scores and the semi-synthetic calibration. The first uses free data and a direct application, the second attacks the assumption that is most plainly violated in financial data, and the third is cheap and produces a tool that can accompany any DML study. The price impact idea is the most interesting economically but the least feasible, because credible instruments and clean data are scarce.

12 Learning Framework

12.1 Prerequisite concepts

- Linear regression, the Frisch-Waugh-Lovell theorem and instrumental variables.
- Potential outcomes, unconfoundedness, overlap and the propensity score.
- Asymptotic theory: delta method, central limit theorem, stochastic orders O_p and o_p, empirical processes and Donsker classes.
- Regularisation and learner rates: lasso, random forests, boosting, neural networks, and what a rate $n^{-\varphi}$ means.
- Pathwise (Gateaux) derivatives and efficient influence functions in semiparametric models.

12.2 Core concepts learned

- Regularisation bias and overfitting bias are two different problems with two different cures.
- Neyman orthogonality turns a first-order bias into a second-order, product-form bias.
- Cross-fitting makes conditional-on-training arguments valid for arbitrary learners and returns full efficiency.
- Nuisance rates matter through a product, so an easy nuisance can compensate for a hard one.
- Reporting uncertainty from the random partition is part of honest inference.

12.3 Important equations

Equation (3) for why the naive estimator fails, Equation (6) for the exact product bias in the partially linear model, Equation (7) for the orthogonality condition, Equation (8) for the general construction, Equation (15) for the main theorem, Equation (18) for double robustness in the average treatment effect, and Equation (19) for the sparse benchmark.

12.4 Methods learned

Construction of orthogonal scores by projection, concentrating out and influence-function adjustment; cross-fitting with K folds and the DML1 and DML2 aggregation; repeated sample splitting with median aggregation; plug-in variance estimators for orthogonal scores; the AIPW and LATE scores; sparse-benchmark calculations for rate conditions.

12.5 Questions to test understanding

COMPREHENSION

1. Why does the naive estimator in Equation (2) fail to be root-n consistent even when $\hat{g}_0$ is consistent?
2. What does Neyman orthogonality require, and why is it useful when $\hat{\eta}_0$ converges slowly?
3. What problem does cross-fitting solve that orthogonality does not?

ANALYSIS

1. Derive Equation (6) from the model, and say which of the conditional mean restrictions is used where.
2. Show that the score $D(Y - D\theta - g(X))$ is not orthogonal and compute its derivative in the direction of g.
3. In Equation (18), which factors become large as the propensity approaches zero, and what does this imply for the rate condition?

RESEARCH

1. How would the proof of Theorem 3.1 change if observations were a stationary mixing sequence and folds were contiguous blocks?
2. Can a diagnostic for the product-rate condition be constructed from data, and what would its power be?
3. What is the right inference for DML when the outcome has an infinite fourth moment?

12.6 Papers to read next

Chernozhukov, Newey and Singh (2022); Belloni, Chernozhukov and Hansen (2014); Chernozhukov, Cinelli, Newey, Sharma and Syrgkanis (2022); Kennedy (2023); Robins and Rotnitzky (1995); Newey (1994).

13 Conclusion

13.1 One-sentence thesis

Combining a Neyman-orthogonal score with cross-fitting makes root-N inference on a low-dimensional parameter valid with essentially any machine learner whose nuisance errors multiply to $o(N^{-1/2})$.

13.2 Three most important insights

- The bias of an orthogonal estimator equals the covariance of two nuisance errors, so inference needs the product of the errors to be small, not each error.
- Overfitting bias is removed by never evaluating a learner on the observations it was trained on, whatever the learner is.
- The recipe is general: any moment model can be given an orthogonal score by projection, concentrating out or an influence-function adjustment.

13.3 Strongest evidence

The theoretical results and their transparent proofs, backed by the exact product identities in Equations (6) and (18), and the reviewer's simulation, in which a naive plug-in estimator is displaced by 17 per cent of the true effect while DML is centred.

13.4 Biggest weakness

The rate condition on the product of nuisance errors is neither tested nor diagnosed, so the validity of the intervals in each application rests on an unverified assumption, and the unconfoundedness assumption is untouched by the method.

13.5 Most important research gap

Inference validity under dependent and heavy-tailed data, the two features that dominate financial applications.

13.6 Best research opportunity

A blocked cross-fitting study of stock characteristics using public data, with a simulation of the size distortion of random-fold cross-fitting under dependence.

13.7 Three papers to read next

Chernozhukov, Newey and Singh (2022); Belloni, Chernozhukov and Hansen (2014); Feng, Giglio and Xiu (2020).

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